

CSE320 (Data Comm) – Practice Sheet on Chapter 2

Q1

Identify the name of the **TCP/IP model** layers based on the following functionalities.

- Enables resource sharing and remote file access among network users. [Application]
- Responsible for converting data into signals for transmission over a physical medium. [Network Access]
- Responsible for establishing and terminating communication sessions. [Application]
- Ensure reliable hop to hop transmission. [Network Access]

Q2

Label the frames (1 & 2) shown below in Figure No. 1 with appropriate port, IP and MAC addresses. The sender Host A has two applications running; one for FTP with port number 51024 and the other for accessing the web server with port number 52348. Frame 1 is coming from the TEST web server to HOST A and Frame 2 is intended for the web server from Host A. Web servers use port number 80 to send data to clients. (For indicating MAC addresses just mention the device or device interface).

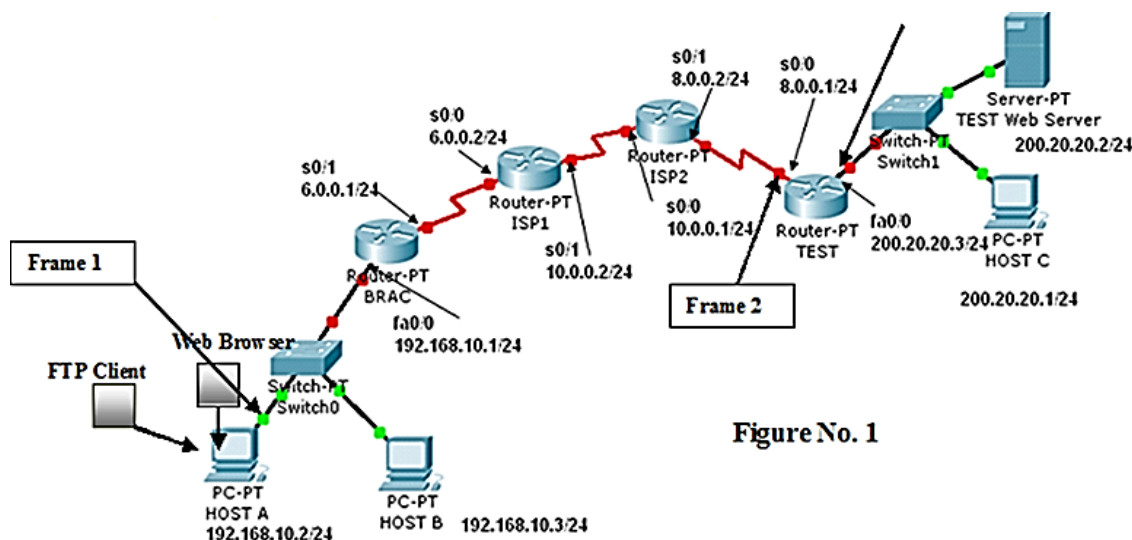


Figure No. 1

Frame 1

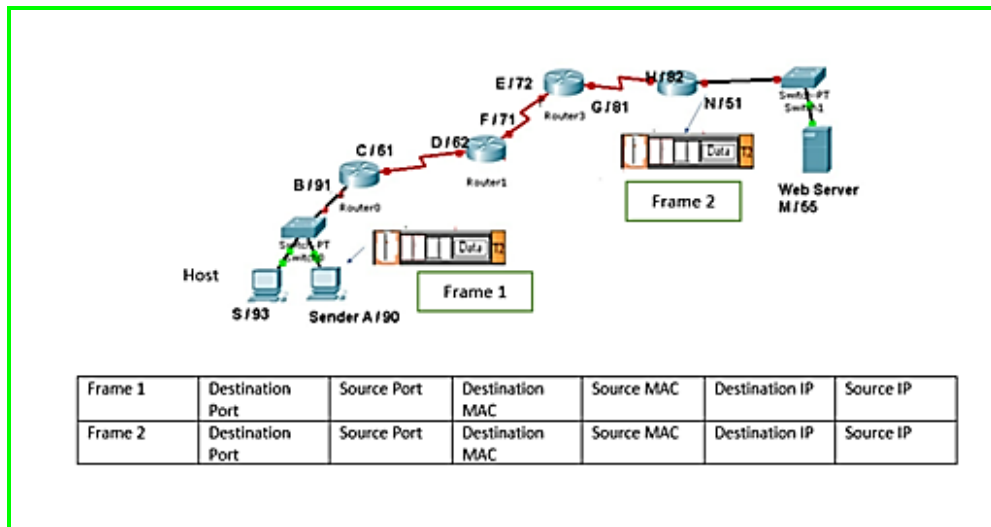
D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
HOST A	Router -PT BRAC	192.168.10.2/24	200.20.20.2/24	52348	80

Frame 2

D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
Router-PT Test	Router-PT ISP2	200.20.20.2/24	192.168.10.2/24	80	52348

Q3

Label the frames (1 and 2) shown below with appropriate Port, IP and MAC addresses. The sender A is requesting for a web page to the Web server by sending frame 1. The web server is sending frame 2 to Host S. Assume the numbers mentioned in the figure represent IP addresses and the alphabets are MAC addresses. While mentioning the port addresses, you can use any value from the appropriate ranges.



Frame 1

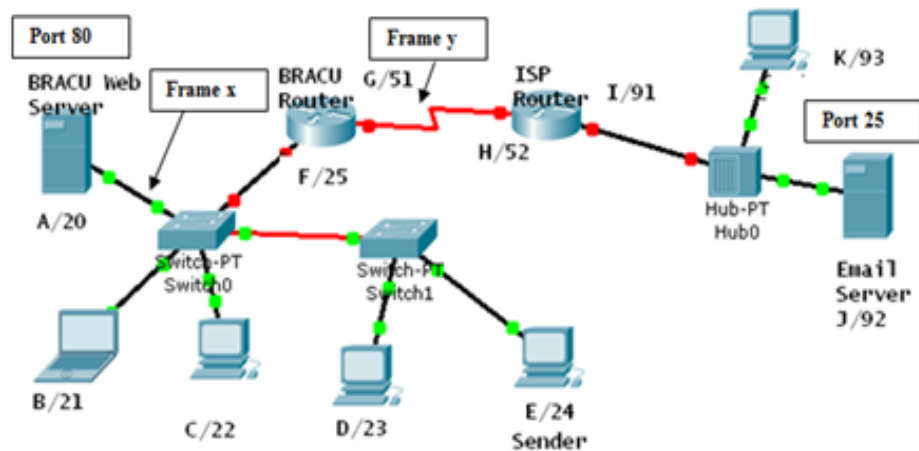
D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
B	A	65	90	80	52148

Frame 2

D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
S	B	93	65	52150	80

Q4.

Complete the frames (X & Y) given below with appropriate port, IP and MAC addresses. The sender Host E has two applications running; one for email with port number 57150 and the other for accessing the web server with port number 52044. The frame X is intended for the BRACU Web server and frame Y is coming from the Email Server. (MAC addresses are alphabets and IP addresses are numbers).



Frame X

D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
A	E	20	24	80	52044

Frame Y

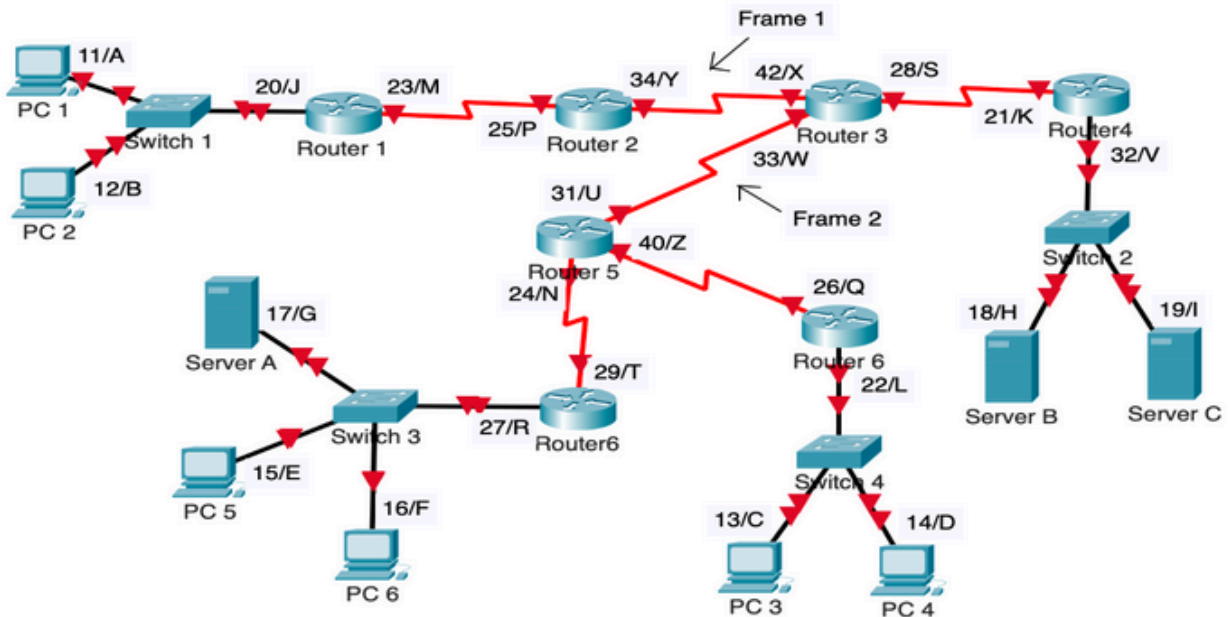
D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
G	H	24	92	57150	25

Q5.

Write the name of the TCP/IP model layers based on the following functionalities

- The layer processes information which helps us identify a network device. [Network]
- The layer responsible for transmitting data over fiber-optic. [Network Access]
- This layer translates messages from one language to another.[Application]
- User Datagram Protocol is used in this layer. [Transport]

Q6



Consider the network topology shown above. Assume the physical addresses to be the uppercase letters and the logical addresses to be the numbers. For the Devices, use a Port number from the dynamic range (49152 - 65535).

- . How many networks are there in total? **Ans: 10**
- i. Data is coming from Server A which uses port number 23 and it is intended for PC2. Considering this situation, **complete** Frame 1 below by writing the appropriate Destination and Source Physical, Logical and Port Addresses.

D.MAC	S.MAC	D.IP	S.IP	D.Port	S.Port
Y	X	12	17	49152	23

- ii. PC5 is trying to send data to Server C, which is a Web Server using port 80. Considering this situation, **complete** Frame 2 below by writing the appropriate Destination and Source Physical, Logical and Port addresses.

Frame 2	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port	Data	Trailer
	W	Z	19	15	80	49513		

Q7.

Identify the name of the TCP/IP model layers based on the following functionalities:

- The layer processes information which helps us identify a network. [Network]
- The layer responsible for transmitting data over fiber-optic. [Network Access]
- This layer translates messages from one language to another. [Application]
- This layer uses sequence numbers to sort segments. [Transport]

Q8.

Charlie is working remotely and needs to securely connect to his company's database to update client information. He begins by initiating a connection that verifies his login credentials. Once verified, the connection must remain stable to prevent any disruptions during his session. As Charlie enters his updates, the software on his computer first ensures that the data is in the correct format and is encrypted before it's sent. The data is then broken down into manageable pieces that are each checked for data integrity before and after transmission. This process also ensures that all data arrives in the exact order it was sent and checks for any duplications or data losses. Additionally, the network manages the path of the data, adapting to network traffic and connectivity changes to ensure efficient routing.

Identify the OSI model layers involved when:

1. Charlie's connection is authenticated and maintained. [Session]
2. Charlie's data is formatted and encrypted. [Presentation]
3. Data is segmented, checked for integrity, and ordered during transmission. [Transport]